There is a countable unconfined Steiner triple system that is not free

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Editorial note. The counted statement is the author's environment at source0.tex lines 1097-1098 and its proof environment opens at line 1099; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

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Editorial note. r002 note: the retained author proof cites the key barcas2, which the article bibliography bundled with the source carries; r002 bundles that bibliography file and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: a countable unconfined system that is not free (unlabelled proposition at source0.tex lines 1097-1098).

Allowed standard prerequisites (editorial).

- The free system on three generators. F_3 is freely generated by three elements a_1, a_2, a_3.. It is the starting point of the construction.

- The iterated union. N_omega is the union of a countable chain of copies of the free system, each elementary in the next.. It is the system shown to be unconfined but not free.
- Freely generated systems. A system is free when it is freely generated by some set, in particular it is generated by that set.. It is what the constructed system fails to be.

Proposition 0.1 Let M, N be models of the theory of free STSs. If $M \leq N$, then $M \preccurlyeq N$.

Proposition 0.2 If $3 \leq n < m$, there is no elementary embedding of F_m, the free STS with m generators, into F_n, the free STS of n generators, despite the elementary equivalence of F_m and F_n.

Proposition 0.3 There is a (countable) unconfined STS which is not free.

Proof. Let $N = F_3$ be the free STS generated by three elements a_1, a_2, a_3, and let
$b = (a_1 \cdot a_2) \cdot (a_1 \cdot a_3)$. As shown in [1], Example 6.9, the homomorphism $f : N \to N$ determined by $f(a_1) = b$, $f(a_2) = a_2$ and $f(a_3) = a_3$ is an embedding but it is not surjective. Hence, there is some proper substructure of N isomorphic to N. So we can define an ascending chain $(N_n \mid n < \omega)$ of STSs such that $N_n \cong N$ and $N_n \neq N_{n+1}$. Let $N_\omega = \bigcup_{n<\omega} N_n$. Since N_ω is a union of a countable chain of unconfined STSs and unconfinedness fails on a finite subset, we have that N_ω is unconfined. Moreover, it is not finitely generated. Since it is countable, if it is free then it has a free base B of ω elements and it is (isomorphic to) the generic. There is an embedding $f : F_4 \to N_\omega$ such that $f(F_4) \leq N_\omega$. Choose n such that the image in f of the base of F_4 is contained in N_n. Then $f(F_4) \leq N_n$ and by Proposition 0.1, $f : F_4 \to N_n$ is elementary. Since $N_n \cong F_3$, this contradicts Proposition 0.2. Hence, N_ω is not free. □

References

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[3] B. Ganter and H. Werner. Equational classes of Steiner systems. Algebra Universalis, 5:125–140, 1975.

[4] B. Ganter and H. Werner. Co-ordinatizing Steiner systems. In C.C. Linder and A. Rosa, editors, Topics on Steiner systems, pages 3–24. North Holland, 1980.